



Project no: 963530.



LEAP-RE

Europe-Africa Research and Innovation call on Renewable Energy LEAP-RE

Full proposal application form

The project description cannot exceed a 30-page limit

1. Project overview

Proposal full title	Recycling of the cathodes, based on carbon nanotubes and conducting polymers, from spent rechargeable Li batteries		
Proposal acronym/short name	RCLIB/Recycling of the cathodes from spent rechargeable Li batteries		
Project Coordinator (organisation + name of the project coordinator)	National institute of Materials Physics (NIMP), Mihaela Baibarac		
Start date ¹ (month/year)	July 1, 2023	Total Project Costs in EUR:	837.600
End date (month/year)	June 30, 2025	Requested funds in EUR:	600.000
Project duration in months			
Topic and sub-topic ²	Topic: (tick the relevant topics(s)) ☐ Topic 1: Mapping joint research and innovation actions for next-step development of RES and integration of RES in sustainable energy scenarios ☒ Topic 2: End-of-life and second-life management and environmental impact of RE components ☐ Topic 3: Smart stand-alone systems ☐ Topic 4: Smart grid (different scales) for off grid application ☐ Topic 5: Processes and appliances for productive uses (agriculture and industry) ☐ Topic 6: Innovative solutions for priority domestic uses (clean cooking and cold chain)		
Supplementary key words	(max. 5) rechargeable Li batteries, conducting polymers, carbon nanotubes, supercapacitors, recycling		

N.B.: Questions and comments in blue may be overwritten or deleted. Please stick to the following formatting requirements: Calibri, 11pt, single spaced for all text except references and footnotes; the pages' margins 2.5 cm should be kept. Both formatting requirements as well as the maximum number of pages (specified in the respective chapters) help the International Review Panel to read and compare the proposals. Please note that you will be asked by the Call Secretariat to shorten your proposal after submission if the intended page limit is exceeded.

¹ Before completing, each partner should inquire the respective Funding Organization.

² Select the main topic addressed by the proposal and one, or more, sub-topics.



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2. Abstract

Development in the field of rechargeable Li batteries (RLIBs) and supercapacitors (SCs) has seen rapid growth as a consequence of the demand for electric vehicles and of various devices such as wind turbines, hybrid energy storage systems for the installation of photovoltaic devices, etc. However, end-of-life (EoL) battery components induce a new environmental challenge, requiring their collection, disposal and recycling/reuse to avoid further environmental pollution. The innovation in this project envisages the life cycle analysis of novel emerging cathodes containing carbon nanotubes (CNs) & conducting polymers (CPs) from RLIB and the development of new ways for the recycling of the constituents of the cathodes at EoL. The main objectives of this project are: a) the collecting of spent RLIB, having the CP/CNs composites as electrode materials; b) the recovery of the cathode and the separation of spent electrode constituents, i.e. the CP_s/CNs composite (CP_s is CP in semi-doped state), polyvinylidene fluoride (PVDF), CN⁻Li⁺, CP in undoped state (CP_u), LiCl, LiPF₆ and super P carbon (C); c) the development of various strategies for improvement of physico-chemical properties of the CP/CNs/PVDF, CP/CNs and CP/oxidized CN (OCN) composites, e.g. by the protonic acid doping of CP, the chemical polymerization of monomers in the presence of CN, OCN, and the fibers resulted by the electrospinning; the first two methods will allow to transform CPs or CP_u into CP in doped state (CP_d); d) synthesis of the CP/Biochar/CNs composites and their characterization; and e) the applications of the prepared composites as electrodes for SC cells & RLIB using both classical methods and electrospinning and comparison of the obtained performance. The methodology and working plan involve the following stages: a) charge/discharge galvanostatic tests of RLIBs to ensure that the cathode based on the CP/CNs composite is at EoL and subsequent its collection; b) separation of the spent electrode constituents, i.e. CPs, PVDF, CNs and super P C, by chemical methods and their optical and structural characterization; c) preparation of new electrodes from CP, PVDF, CNs & super P C, and their testing in the SC cells & RLIB; and d) optimization of SCs & RLIBs, using prepared composites from the cathodes of spent RLIBs. To implement this methodology, we will employ: a) charge/discharge galvanostatic measurements, b) cyclic voltammetry; c) electrochemical impedance spectroscopy; d) Raman scattering, e) IR spectroscopy, f) X-ray photoelectron spectroscopy, g) X-ray diffraction, h) scanning electron microscopy, j) transmission electron microscopy, k) contact angles, l) thermogravimetry, etc. The stakeholder involvement will be: a) NIMP – the preparation of new composites, resulting from the chemical polymerization of monomers in the presence of CN & OCN, their characterization and the electrospun fibers as well as testing in SC cells & RLIB; b) CITST - electrospun fibers from CP/CN/PVDF blends and exploitation aspects; c) UM6P, UCA & UBMA - testing of the electrodes containing CP_d resulted by the protonic acid doping and their hybrids, in RLIB & SC cells; and d) INFN - testing of composites, resulted by chemical polymerization of monomers in the presence of CN and OCN as well as the fibers. The expected impact consists of: a) development of a new technology for the separation of the spent cathodes constituents at EoL; b) access to recyclable technology with impact on environment, for the beneficiaries from AU and EU ; c) development of new SC cells & RLIB, using recycled compounds from spent RLIB; c) formation of new experts in the field of energy storage and wastes management to aid the transition from linear to circular economy; d) the knowledge/technological transfer, which will strengthen the cooperation between the participating institutions, having a direct benefit on the energy security of each country, enabling waste management in the field of energy storage.



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3. Coordinator/partner profile

	Organisation	Type of organisation ³	Country / Funding agency ⁴	Contact Person (first name and family name)	Gender (Female/ Male/ Non-binary)
Project Coordinator/Main Applicant	National Institute of Materials Physics (NIMP)	RO	Romania/ UEFISCDI	Mihaela Baibarac	Female
Project Partner 2	IT Center for Science and Technology (CITST)	SE	Romania/ UEFISCDI	Oana Cramariuc	Female
Project Partner 3	Istituto Nazionale di Fisica Nucleare (INFN)	RO	Italy/MuR	Stefano Bellucci	Male
Project Partner 4	Mohammed VI Polytechnic University (UM6P)	RO	Morocco/ MENFPESRS	Saadoune Ismael	Male
Project Partner 5	Badji Mokhtar University (UBMA)	RO	Algeria/ MESRS	Ouanassa Guellati	Female
Project Partner 6	University Cadi Ayyad (UCA)	RO	Morocco/ MENFPESRS	Rachid Hakkou	Male

Please insert additional rows if necessary: place cursor in last row and select "Table → insert → row below" from the menu bar.

³ Type of organisations: SE = small enterprise; ME = medium-sized enterprise; LE = large enterprise; RO = research organisation, OTH = other type of organization. *Note: With regard to the size of companies, for all EU member states the current definitions of SMEs given in the EU competition law are applied (definition of small and medium-sized enterprises and of independent businesses in accordance with recommendation 2003/361/EC of the Commission dated 6 May 2003, [ABl. L 124 of 20.5.2003, pp. 36-41]; cf. http://ec.europa.eu/enterprise/policies/sme/files/sme_definition/sme_user_guide_en.pdf).*

⁴ If applicable. For partners from African countries without funding agency asking for funding, please indicate here with the funding ask



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4. Technical description of the project

a) Scientific and/or technological excellence (max. 20 pages)

Project objectives and targets

Development in the field of rechargeable Li batteries (RLIBs) and supercapacitors (SCs) has experienced unprecedented growth as a consequence of the increasing demand for electric vehicles and of various devices such as wind turbines, hybrid energy storage systems for the installation of photovoltaic devices, etc. However, end-of-life (EoL) battery components induce a new environmental challenge, requiring their collection, disposal, and recycling/reuse to avoid further environmental pollution, preserve the environment, and access critical natural resources. Consequently, new energy paradigms are emerging in both Africa and Europe in recognition of the need to develop the EoL management of storage products without delay. In this respect, the focus lies on conceiving and fabricating functional low-cost materials from EoL components with the aim of integrating them into key electronic components for societal and industrial development. With this background in the mind, we propose recycling cathodes from spent RLIB with a specific capacity equal to 157.2 mF/cm^2 [1]. The cathode of the spent rechargeable Li batteries (RLIBs) of the type (-) Li/ conducting polymer/carbon nanotubes (CP/CNs) composite (+) shows the following composition: the composite based on CP in semi-doped state (CP_s)/CN, CP in un-doped state (CP_u), CN doped with Li^+ ions (CN^-Li^+), polyvinylidene fluoride (PVDF), super P carbon, LiPF_6 , and LiCl , where CP is poly(diphenylamine) (PDPA) [1]. The **overall objective** of this project is the recycling of the cathodes, from spent RLIBs of the type (-) Li/ the PDPA/CNs composite (+), by the fabrication of new electrodes containing the CP_d /CNs, CP_d /OCN, CP_d /CNs/PVDF & CP_d /Biochar/ CNs composites (CP_d and OCN stand for CP in doped state and oxidized CN, respectively) as active materials for supercapacitor (SC) cells and RLIB. The **specific objectives** of the project are: a) the development of a technology to separate the constituents of the cathodes from spent RLIBs to prepare new electrodes for SC cells and RLIBs; b) the preparation of the CP_d /CNs, CP_d /OCN, CP_d /CN/PVDF & CP_d /Biochar/CNs composites and the optimization of electrodes containing above composites for SCs, to reach the discharge capacitance values comparable with those reported for the cells which use SWNTs functionalized with PDPA doped with $\text{H}_3\text{PW}_{12}\text{O}_{40}$ anions as electrode materials, i.e. 157.2 mF/cm^2 [1]; and c) usage of both conventional (e.g. in-situ polymerization, etc.) and unconventional methods (electrospinning) for the optimization of electrodes containing the CP_d /CN composites for RLIB, so that the value of the specific capacity of these batteries becomes comparable with that reported in the case of battery (-) Li/the PDPA-salt/SWNT composite (+), i.e. 245 mA hg^{-1} [2]. A schematic presentation of the specific objectives and the scope of the multiannual roadmaps (MARs) addressed are shown in Table 1, where both the ways in which the specific objectives are addressed in the project and the expected outcomes are highlighted, all this information allowing the estimation of the impact presented in the Impact section b.

Table 1. MAR's objectives mapped with the project implementation and outcome

MAR	Objectives	Methods used in the RCLIB project	Outcome
2	The development of a technology to separate the constituents of the cathodes from spent RLIBs	This technology involves cathode disassembly steps and separations of the constituents by physico-chemical processes –Task 1.3	Technology can be easily deployed both in EU and AU countries.
2	New methods for preparation of the active materials for new energy storage devices using the constituents from spent RLIBs	Adapted synthesis of composites from constituents of spent RLIB, which will allow the transition from a linear to a circular economy – Tasks 2.1, 2.2, 2.3	The development of materials without the environmental degradation



2	<i>The optimization of electrodes containing the CP_a/CNS, CP_a/OCN & CP_a/CN/PVDF composites and their testing in SCs and RLIBs</i>	<i>Adapted electrode preparation methods and the testing of electrodes in SCs and RLIBs – Tasks 3.1, 3.2</i>	<i>Sustainability of the energy storage systems</i>
2	<i>Assessment of needs and resources at country and/or regional level</i>	<i>Task 6.1 will identify key stakeholders for future commercialization through EU-AU cooperation and will assess needs and user satisfaction, both in the case of CITST and SMEs which have sent the support letters for this project, i.e. Qi technologies and RAIT88 SRL, as well as other SMEs identified during the project implementation</i>	<i>Identification of key R&I partners for the collaboration in future joint initiatives</i>
2	<i>Progress of the cooperation between partners of the countries from EU and AU</i>	<i>This project supports cooperation between EU and AU during the implementation, by finding of new partners for waste management from energy storage devices and commercialization of the products obtained in this project.</i>	<i>Long-term cooperation between countries from EU and AU</i>

The main goals are presented in the Key Performance indicators (KPIs) in Table 2, which take into account the qualitative KPIs in the ratio to the functional and non-functional requirements that will reflect the satisfaction of the end-users, these being revised during the project implementation.

Table 2. The main KPIs of the RCLIB project

KPI description	KPI target	Means of verification
<i>Full-scalable synthesis of composite materials from constituents of the cathodes of spent RLIB</i>	<i>Four composites of the type: CP_a/CNS, CP_a/OCN, CP_a/CNS/PVDF & CP_a/Biochar/CNS; 10 g from each composite</i>	<i>Synthesis and product sheets with the main physical properties (Raman spectra, IR spectra, XPS spectra, etc.) will be performed in the labs from EU and AU (Tasks 2.1, 2.2, 2.3, 2.4)</i>
<i>The discharge capacitance value of SCs</i>	<i>157.2 mF/cm²</i>	<i>Measurements performed in the labs from EU and AU (Task 4.1)</i>
<i>The specific capacity of RLIBs</i>	<i>245 mA hg⁻¹</i>	<i>Measurements performed in the labs from EU and AU (Task 4.2)</i>
<i>Concentration of active material in the electrodes of SCs & RLIBs</i>	<i><85 wt.%</i>	<i>Measurements performed in the labs from EU and AU (Tasks 4.1 and 4.2)</i>
<i>Stakeholder satisfaction of the electrodes having the active composite materials in the SCs and LIBs</i>	<i>> 85%</i>	<i>Based on the results obtained in the Task 6.1</i>

The main KIPs of the RCLIB project will take into account the design centered on future stakeholders (e.g. companies which provided support letters), when an optimization of: a) the technology of the separation of constituents from the cathodes of spent RLIB; b) the synthesis of composites from the constituents of the spent RLIB cathodes; and c) the electrodes used in SCs and RLIBs in the term of the



performance of these energy storage devices, so that their cost is competitive with existing commercial products.

Current state of art and progress beyond the state-of-the-art

Among different clean energy technologies, LIB is one of the most important products being a major power source in consumer electronics and electric vehicles (EVs). The rapid expansion of the LIB market amplifies the potential risks in its upstream supply chain, which consists of many “critical materials”, e.g., lithium, cobalt, nickel, and graphite [3]. These materials are deemed to be critical because of their high supply concentration, significant environmental and social impacts, high importance to the performance of final products, low substitutability, and other factors. While lithium is mostly recycled via hydrometallurgical technologies and is not the focus of our project, the main strategies adopted for recyclable wastes are: a) the spent battery powder, the oxidative polymerization of aniline and pyrrole in the presence of graphite, when composites of the type polyaniline/graphite and polypyrrole/graphite were obtained [4, 5]; b) single-walled carbon nanotubes (SWNTs), b₁) the reflux of CNs in the presence of HNO₃ accompanied of the thermal oxidation in air or b₂) the interaction with HCl and the thermal oxidation [6] and c) the decomposition of PVDF from the cathode of spent RLIB [7]. In this project, using prepared electrodes according a previous protocol proposed for RLIB by the NIMP’s team [2], we go one step further and propose a novel technology for recycling the spent RLIBs cathodes. Furthermore, we demonstrate that the separation of the cathode constituents from spent RLIB, allows the preparation of new composites which can be used in RLIBs & SCs. A comparison of the performance of the CP_d/CNs composites with those synthesized in the presence of Biochar [8] in SCs is also envisaged. In addition, we will employ **electrospinning** for the preparation of fibers from the above composites. Electrospun electrode materials were proven to hold several promising properties due to their high surface area, high porosity, tunable surface properties, and low cost, among others [9,10]. At the same time, modern material science often makes use of PVDF thin films because of various properties, like high thermal and chemical stability. Fibers of this polymer material are, on the other hand, much less explored. By introducing CNs via electrospinning, it is possible to affect the chemical and electrical properties of the resulting composite [11]. In this project, we are exploiting all these features while advancing beyond current state-of the art (SoTA) by considering various doped CN and their influence on the cathode properties. Also, CNs are a candidate material for use in LIB due to their unique properties, e.g. electronic conductivity, large specific surface area, favorable chemical stability & low-dimensional structure [12]. According to the IDTechEx report “Carbon Nanotubes 2022-2032: Market, Technology, Players”, CNs usage is expected to be of up to 70 kilograms tons per year in the next 10 years [13]. The incorporation of CNs as a conductive additive at a lower weight loading than conventional carbons, like carbon black and graphite, presents a more effective strategy to establish an electrical percolation network. In addition, CNTs have the capability to be assembled into free-standing electrodes (absence of any binder or current collector) as an active Li⁺ ion storage material or as a physical support for ultra-high capacity anode materials. The approach of the RCLIB project is highlighted in Table 3, where the difference compared to the current SOTA methods is highlighted.

Table 3. State-of-the-art of material recycling from spent LIBs and progress proposed in RCLIB

Process stage	Method	Current SOTA	RCLIB beyond SOTA
LIB pretreatment technologies	Mechanical separation	Separate electrode materials encapsulated in iron and plastic by crushing, grinding and/or magnetic separation [14]	Separation of cathode by mechanical dismantling of RLIB, followed of: a) the chemical interaction of



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	<i>Thermal treatment</i>	<i>Organic additives and binders are pyrolyzed at high temperatures to liberate electrode materials from foils [15]</i>	<i>cathode with H₂O, in order to eliminate the salts onto the cathode surface, b) the successive interaction with N-methyl-pyrrolidone (NMP) when results a solution of the PVDF binder and CP/CNs composite and a solid fraction of super P carbon and CNs. The filtration process will allow the recovery of: a) solution of PVDF binder and CP/CN composite and b) the mixture consisting in super P carbon and CNs. The separation of super P carbon and CNs will be carried out by centrifugation</i>
	<i>Mechano-chemical treatment</i>	<i>The use of CaO as the reaction medium for the decomposition of the PVDF in the cathode electrode can be achieved by the thermat treatment at 300 °C, without the release of hydrogen fluoride [16]</i>	
<i>Metal element extraction</i>	<i>Pyrometallurgical process</i>	<i>Pretreated LIBs are burned in the smelter, where organic materials (plastic and the separator) will be burned away. The metals can be converted to alloys (e.g., Co, Fe-, and Ni-based metal products) through carbon reduction in a high-temperature environment. Li, the critical element, remains in the slag fraction and can be recovered with extra processes [17].</i>	<i>After pretreatment, the collected spent cathode materials will be regenerated by restoring the crystal structure and simultaneously offsetting the Li loss of electroactive in situ, since the root of battery performance decay is generally attributed to inactive Li supply and irreversible structural transition.</i>
	<i>Hydrometallurgical process</i>	<i>Solid metal elements are converted into soluble substances through various leaching processes (acid-, alkali, and bio-leaching) [18].</i>	

According to Table 3, this project will unlock an area of the economy in both the EU and AU that will allow the introduction of lower cost storage options and a sustainable life cycle that will not end with thermal disposal or landfilling hazardous materials. Overall, research in the EoL LIBs has increased sharply in recent years with more than 35 projects funded under Horizon 2020, i.e. LithoRec and LithoRec2, Batteries 2020, ProSum, etc.[19]. The Table 4 below is presenting some of these projects. These mostly concentrate on the recovery of rare metals.

Table 4. The main objectives of funded projects

Funded projects	Objectives
<i>LithoRec and LithoRec2</i>	<i>Develop and scale up a complete recycling process for electric car batteries. The method is based on a combination of thermal, mechanical and hydrometallurgical processes where the result has become a process with a very high recycling rate of most materials in several different chemistry.</i>
<i>Batteries 2020</i>	<i>Has improved cathode materials based on nickel/manganese/cobalt oxides. Such materials have a high chance to be up-scaled and commercialized near-term.</i>



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AutoBatRec2020	Aims to recycle automotive batteries which contain strategic elements such as cobalt, graphite, lithium, nickel that are not readily available in Europe. EoL batteries will be valorized, while conforming with environmental directives, to reduce the European dependency in critical materials.
SIREVIVAL	Aims at the reduction of the environmental impact of spent photovoltaic modules. The effort is focused using recycled materials, like Si, to build all solid-state supercapacitors.

Originality and/or innovation of the concept and approach (indicate start and target TRL5, if applicable)

In the RCLIB project proposal, the technology for the recyclable of the cathodes from spent RLIB involves: a) the recovery of the cathode and the elimination of inorganic salts coming from the electrolyte of the spent RLIB, b) the separation of the cathode constituents by the interaction of the spent electrode with N-methyl pyrrolidone (NMP), when the mixture of the CP_s/CN composites, CP_u and PVDF are separated of CNs and super P carbon; c) the interaction of the mixture of the CP_s/CN composites, CP_u and PVDF with alkaline solutions leads to a the CP_u/CN, CP_u, PVDF blend and OCN (CNs which have grafted the carbonyl and carboxyl groups) [2,20]; in order to use these compounds the CP_u/CN, CP_u and PVDF mixture is exposed to a protonic acid doping resulted in the CP_d/CN, CP_d and PVDF blend, while OCN is used in the radical polymerization of diphenylamine (DPA) to obtain the CP_d/OCN composite; this project will use the CP_d/CN, CP_d and PVDF blends to produce and characterize electrospun electrodes exhibiting large area and extraordinary properties conferred by the micro- and nano-sized fibers [21]; d) the centrifugation processes of CNs and super P carbon will allow to separate of the two constituents, CNs being used afterwards in the chemical polymerization of diphenylamine (DPA) to obtain the CP_d/CN composites; the adding Biochar at the above polymerization reaction will lead to the CP_d/Biochar/CN composites; and e) all composites/hybrid materials will be used to prepare the electrodes for SCs and RLIBs. A schematic representation of these processes is shown in Figure 1.

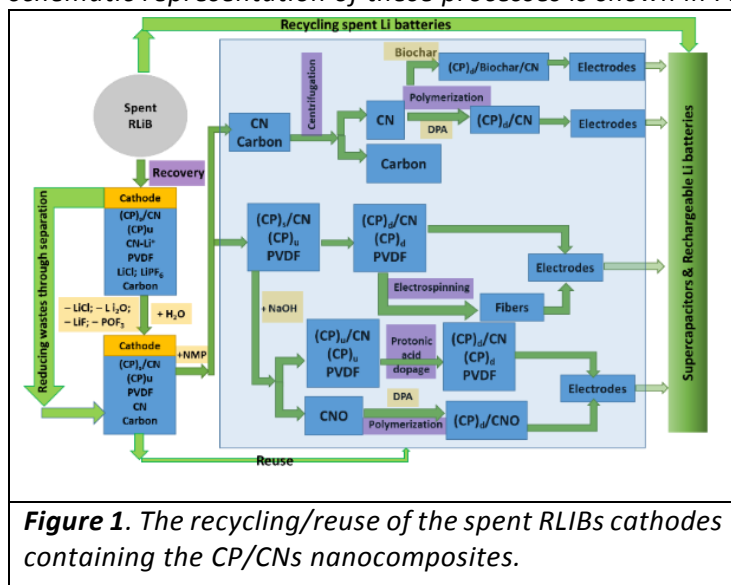


Figure 1. The recycling/reuse of the spent RLIBs cathodes containing the CP/CNs nanocomposites.

The originality and/or innovation of the concept and used approach for recycling of the PDPA-PVDF blend, MWNTs and super P C consist in: a) the separation technology of spent cathode constituents; b) the reuse of the CP/CNs, CP & PVDF blend, after a protonic acid doping process, as active materials in electrodes of RLIBs and SC cells; c) the reuse of CN & OCN by their addition to a mixture of the radical polymerization reaction of DPA when new composites/hybrids result, these being used afterwards as electrodes in RLIB & SC cells; d) fabrication of the CP/CNs/PVDF fibers by

electrospinning processes in order to prepare electrodes for RLIB and/or SC cells as outlined in the previous section. The project starts from previous developments (TRL 2 level) related to: the synthesis, the optical & structural characterization of the CP/CNs composites (NIMP, CITST, UBMA, INFN), the fibers preparation by electrospinning (CITST), the electrode design (UM6P, UBMA, NIMP, INFN, UCA), the applications in RLIBs and SCs (UM6P, UBMA, NIMP, INFN, UCA) and

⁵ TRL – Technology Readiness Level. H2020 definition available in Appendix II.



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*aims to integrate the wastes management from the energy storage field in an unitary concept of TRL 4 and finally achieve an experimental model of RLIB and SC with the recycled/reused wastes, which are to be technology validated in relevant conditions by teams from Morocco and Algeria, thus targeting **TRL5**.*

Methodology

i) Describe the overall strategy of the work plan (max. 1 page)

The RCLIB project will be implemented by a trans-disciplinary (industrial and academic) consortium with experts from Europe, Mediterranean Partner Countries in the fields of nanotechnology, energy storage and waste recycling. In this context, the expertise of the partner teams is in the field of synthesis / characterization of different kind of organic/macromolecular compound/ inorganic nanomaterials (1D, 2D, 3D) [22-32], monitoring and assessment of organic & inorganic contaminants and environmental impact assessment and energy storage system technology development. Beyond producing relevant knowledge on technologies and impacts in key areas of energy storage, RCLIB will perform an environmental benefit analysis and socio-economic impacts (WP5) to make accessible both existing knowledge and results from novel, innovative research and development on recycling technologies of cathodes from spent RLIB for future energy storage devices, considering the business plan and exploitation strategies. In key areas of energy storage technology (SCs & RLIBs), we will assess environmental impacts of the methods by performing recycling/recovering of the cathodes and the use of constituent resulted from spent cathodes in the synthesis of different kind of electroactive nanocomposites/nanohybrids for the energy storage devices field as laboratory tests. For the development of the technology of recycling/reuse of the spent RLIBs cathodes containing the CP/CNs nanocomposites, the strategy of the work plan of the RCLIB project involves the implementation of the following work packages (WPs): **WP1: Recovery of the cathodes from spent RLIB and separation of the spent electrode constituents** (months 1-6), coordinator (CO): NIMP; Task 1.1 Synthesis & characterization of the PDPA/CNs composite; Task 1.2 Assembling of RLIB, the assessing of the specific discharge capacity of RLIB at EoL and the collecting of spent RLIB. Task 1.3 Recovery of the cathodes and chemical separation of the PDPA/CNs, PVDF, super P C & OCN; Task 1.4 Optical & structural characterization of constituents of spent cathodes; **WP2: Synthesis of the composites using constituents resulted from spent cathodes** (months 3-11), CO: CITST. Task 2.1 Synthesis of composites: PDPA/CN/PVDF and PDPA/OCN; Task 2.2 The preparation of fibers using PDPA/CN/PVDF; Task 2.3 Synthesis of PDPA/Biochar/CN composite; Task 2.4 Characterization of composites; **WP3: Demonstration of the performance of the composites as electrodes in RLIBs and SCs** (months 9-19) CO: UM6P. Task 3.1 Testing electrodes having the PDPA/CN/PVDF, PDPA/CN, PDPA/OCN & PDPA/Biochar/CN composites as active materials in SCs; Task 3.2 Testing electrodes containing the PDPA/CN/PVDF, PDPA/CN and PDPA/OCN composites in RLIBs; **WP4 Optimization of SCs & RLIBs** (months 18-24), CO: UBMA. Task 4.1 Optimization of electrodes in SC cells; Task 4.2 Optimization of electrodes in RLIB; **WP5 Dissemination of results** (months 1-24) CO: INFN. Task 5.1 Publication of min. 5 ISI articles & 6 contributions at national/ international conference; training for young researchers concerning the composites synthesis & their characterization; Task 5.2 Organization of: a) a kick-off meeting and b) a close meeting; Task 5.3 Workshop in Romania concerning the results obtained; Task 5.4 A project website. **WP6 Technological transfer** (months 18-24) CO: UCA Task 6.1 Business plan for future exploitation; Task 6.2 One patent application. **WP7 Management** (months 1-24); CO: NIMP. Task 7.1 Technical management of project; Task 7.2 "Quality management plan", which will provide documentation & communication procedures, etc.; Task 7.3 "Risk management plan" for to minimize the risks & to implement mitigation action; Task 7.4 Management committee meetings. The technology of recycling/reuse of the spent RLIBs cathodes as well as SCs cells and RLIBs developed within the consortium will be used both in EU (Romania and Italy) and AU (Algeria and Morocco). A business plan for the future exploitation of the results obtained in the RCLIB project will foster long-term cooperation between Europe and Africa. The technology of recycling/reuse of the spent RLIBs



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cathodes will take into account the needs of end-users for recycling/reuse of electrodes from the RLIB and local opportunities in the UE and AU. In order to estimate a price of this technology at laboratory level, using the existing infrastructure, the presently known retail prices will be used.

ii) The timing of the different work packages (WPs) and their components (Gantt chart or similar);

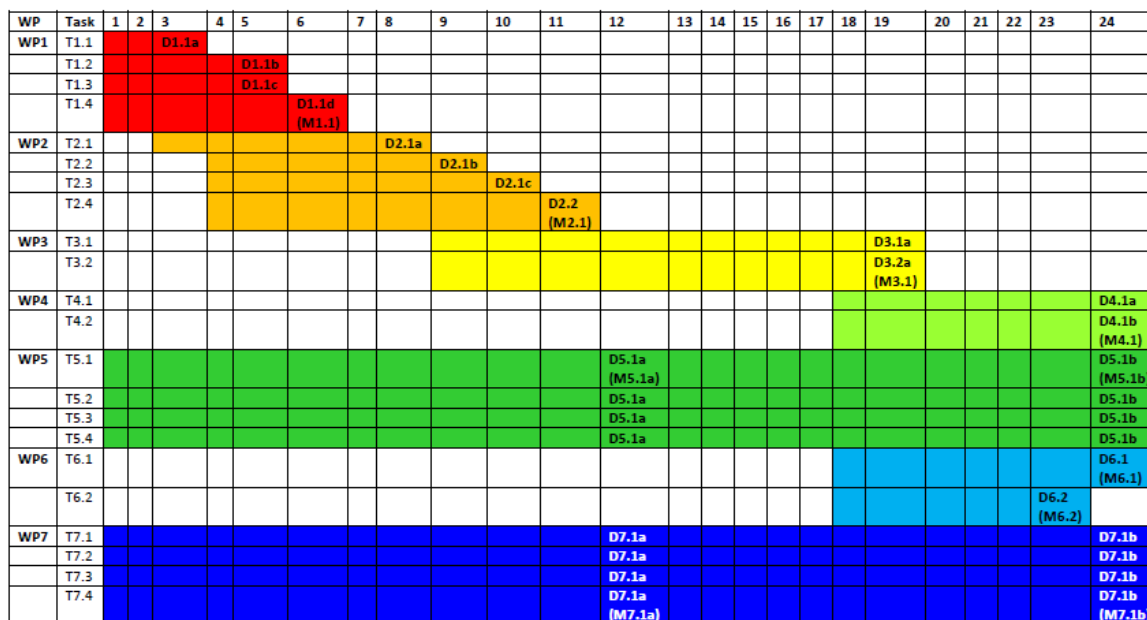


Figure 2: GANTT chart shows the duration of each task (T), planned deliverables (D) and milestones (M).

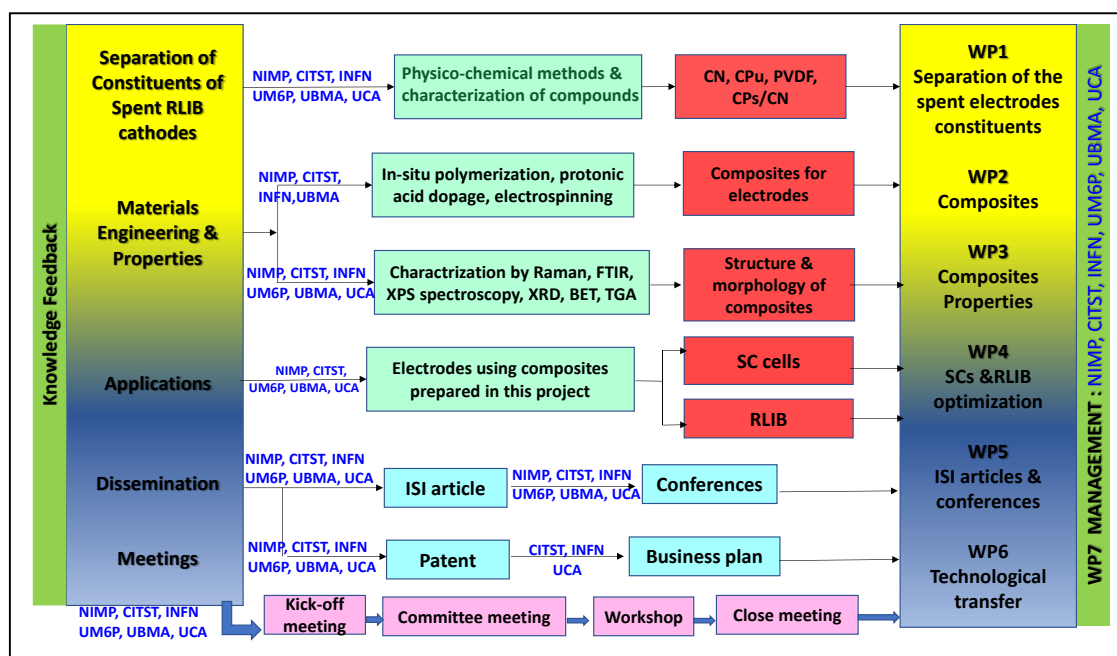


Figure 3. Management scheme for the partnership

iii) Provide a detailed work description broken down into WPs: Work package list (Table 4.1); Deliverables list (Table 4.2); List of milestones (Table 4.3); Description of each WP (Table 4.4, max. 1 page per work package);



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Table 4.1: Work package list

Work package No ⁶	Work package title	Lead project partner No ⁷	Lead project partner short name	Person-months ⁸	Start month ⁹	End month
WP1	<i>Recovery of the cathodes from spent RLIB and separation of the spent electrode constituents</i>	1	NIMP	15	1	6
WP2	<i>Synthesis of the composites using constituents resulted from spent cathodes</i>	2	CITST	21	3	11
WP3	<i>Demonstration of the performance of the composites as electrodes in RLIBs and SCs</i>	4	UM6P	23	9	19
WP4	<i>Optimization of SCs & RLIBs</i>	5	UBMA	16	18	24
WP5	<i>Dissemination of results</i>	3	INFN	24	1	24
WP6	<i>Technological transfer</i>	6	UCA	12	18	24
WP7	<i>Management</i>	1	NIMP	11	1	24
		TOTAL		122		

Table 4.2: Deliverables List

Del. no. ¹⁰	Deliverable name	WP no.	Delivery date ¹¹
D7.1a	<i>Annual report of the project</i>	WP7	M12
D7.1b	<i>Annual report of the project</i>	WP7	M24
D6.1	<i>Business plan for future exploitation</i>	WP6	M24
D6.2	<i>A patent application</i>	WP6	M24
D5.1a	<i>Report on the result dissemination</i>	WP5	M12
D5.1b	<i>Report on the result dissemination</i>	WP5	M24

⁶ Work package number: WP 1 - WP n.⁷ Number of the project partner leading the work in this work package.⁸ The total number of person-months allocated to each work package.⁹ Measured in months from the project start month (month 1).¹⁰ Deliverable numbers in order of delivery dates. Please use the numbering convention <WP number>.<number of deliverable within that WP>. For example, Deliverable 4.2 would be the second deliverable from WP4.¹¹ Measured in months from the project start month (Month 1).



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<i>D4.1a</i>	<i>Report on optimized SCs</i>	<i>WP4</i>	<i>M24</i>
<i>D4.1b</i>	<i>Report on optimized RLIBs</i>	<i>WP4</i>	<i>M24</i>
<i>D3.1a</i>	<i>Report on the electrodes performance in SCs & RLIBs</i>	<i>WP3</i>	<i>M12</i>
<i>D3.1b</i>	<i>Report on the electrodes performance in SCs & RLIBs</i>	<i>WP3</i>	<i>M24</i>
<i>D2.1a</i>	<i>Report on the synthesis of composites: PDPA/CN/PVDF & PDPA/OCN</i>	<i>WP2</i>	<i>M12</i>
<i>D2.1b</i>	<i>Report on the preparation of fibers PDPA/CN/PVDF</i>	<i>WP2</i>	<i>M12</i>
<i>D2.1c</i>	<i>Report on synthesis of composites PDPA/Biochar/CN</i>	<i>WP2</i>	<i>M12</i>
<i>D2.2</i>	<i>Report on the physico-chemical properties of composites</i>	<i>WP2</i>	<i>M12</i>
<i>D1.1a</i>	<i>Report on the properties of the PDPA/CN composite</i>	<i>WP1</i>	<i>M12</i>
<i>D1.1b</i>	<i>Report on the characteristics of spent RLIBs</i>	<i>WP1</i>	<i>M12</i>
<i>D1.1c</i>	<i>Report on the recovery of constituents of spent RLIB cathodes</i>	<i>WP1</i>	<i>M12</i>
<i>D1.1d</i>	<i>Report on the separation of the spent cathodes constituents</i>	<i>WP1</i>	<i>M12</i>

Table 4.3: List of milestones

Milestones are control points where decisions are needed with regard to the next stage of the project. For example, a milestone may occur when a major result has been achieved, if its successful attainment is required for the next phase of work.

Milestone number	Milestone name	Work package(s) involved	Expected date ¹²
<i>M7.1a</i>	<i>Annual report</i>	<i>WP7</i>	<i>M12</i>
<i>M7.1b</i>	<i>Annual report</i>	<i>WP7</i>	<i>M24</i>
<i>M6.1</i>	<i>A business plan</i>	<i>WP6</i>	<i>M24</i>
<i>M6.2</i>	<i>A patent application</i>	<i>WP6</i>	<i>M23</i>
<i>M5.1a</i>	<i>Min. 2 ISI articles & min. 2 contributions at national/international conferences</i>	<i>WP5</i>	<i>M12</i>
<i>M5.1b</i>	<i>Min. 3 ISI articles & min. 4 contributions at national/international conferences</i>	<i>WP5</i>	<i>M24</i>
<i>M4.1</i>	<i>The characteristics of the SCs and RLIBs</i>	<i>WP4</i>	<i>M24</i>
<i>M3.1</i>	<i>The electrode composites materials performance in SCs and RLIBs</i>	<i>WP3</i>	<i>M24</i>
<i>M2.1</i>	<i>The product sheets of composites</i>	<i>WP2</i>	<i>M12</i>
<i>M1.1</i>	<i>The separation of the constituents from spent RLIB cathodes</i>	<i>WP1</i>	<i>M12</i>

¹² Measured in months from the project start month (Month 1)



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Table 4.4: Work package description (for each work package, max. 1 page per work package)

Work package number	1	Start date or starting event: M1				End data: M6	
Work package title	Recovery of the cathodes from spent RLIB and separation of the spent electrode constituents						
Project partner number	1	2	3	4	5	6	Total
Project partner short name	NMP	CITST	INFN	UM6P	UBMA	UCA	
Person-months per applicant:	2	2	3	2	2	4	15
Objectives: <i>The main objective is the recovering of the cathodes, containing PDPA/CNs composite, from spent RLIBs and elaboration of a technology for the separation of the constituents of these electrodes.</i>							
Description of work (possibly broken down into tasks) and role of applicants <i>This work package will be coordinated by NIMP.</i> Task 1.1 <i>Synthesis & characterization of the PDPA/CNs composite – in the frame of this task, we will prepare the composites based on PDPA and CNs by chemical polymerization of diphenylamine (DPA) in the presence of the oxidative media (e.g., K₂Cr₂O₇ & H₂SO₄) and MWNTs as well as the electrochemical polymerization of DPA in the presence of a semi-aqueous solution of HCl in DMF: H₂O (1:1 vol), using cyclic voltammetry (CV) as method [2]. Composite materials will be characterized by Raman scattering, FTIR spectroscopy, X-ray photoelectron spectroscopy, etc. Results will be reported in D1.1a. NIMP will lead this task with participation of all partners.</i> Task 1.2 <i>Assembling of RLIB, the assessing of the specific discharge capacity of RLIB at EoL; the collecting of spent RLIB. This task will involve the preparation of the electrodes based on PDPA/CN using samples prepared in Task 1.1 according to the protocol described in [1] and the assessing of the specific discharge capacity of RLIB at EoL by galvanostatic charge-discharge (GCD) measurements that will be performed in the presence of the electrolyte lithium salt (e.g., LiPF₆) in ethylene carbonate & dimethyl carbonate. Results will be reported in D1.1b. UM6P will lead this task with participation of NIMP.</i> Task 1.3 <i>Recovery of the cathodes and chemical separation of the PDPA/CNs, PVDF, super P C & OCN; - in this task, the cathodes recovered from spent RLIB will be dismutation. Cathodes will be interacted with: a) H₂O for elimination of various salts that there is onto surface of electrodes (e.g., LiPF₆, LiCl, etc.), and b) N-methyl pyrrolidone (NMP) to dissolve the PVDF, (CP)_u and the (CP)_s/CN composite. The separation of super P carbon and CNs will be performed by the centrifugation processes. Results will be reported in D1.1c. UCA will lead this task with participation of all partners.</i> Task 1.4 <i>Optical & structural characterization of constituents of spent cathodes. The mixture of PVDF, (CP)_u and the (CP)_s/CN composite as well as CNs resulted at the end of the centrifugation process will be characterized by Raman scattering, FTIR spectroscopy, X-ray photoelectron spectroscopy, scanning electron microscopy (SEM), etc. Results will be reported in D1.1d. INFN will lead this task with participation of all partners.</i>							
Deliverables (brief description and month of delivery) D1.1a <i>Report on the properties of the PDPA/CN composite will be focused on the morphological, vibrational properties, XPS analysis and thermogravimetry.</i> D1.1b <i>Report on the characteristics of spent RLIBs will show the GCD measurements and the assessing of the specific discharge capacity of RLIB at EoL.</i> D1.1c <i>Report on the recovery of constituents of spent RLIB cathode will show the vibrational properties of cathodes of spent RLIB before to start the protocol of the separation of constituents of these electrodes.</i> D1.1d <i>Report on the separation of the spent electrode constituents will detailed the separation process of the constituents of the spent RLIBs cathodes and the highlighting of the physical properties of the products: a) the mixture of PVDF, (CP)_u and the (CP)_s/CN composite & b) CNs resulted at the end of the centrifugation process. This is a milestone of the project labeled as M1.1.</i>							



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Work package number	2	Start date or starting event: M3				End data: M11	
Work package title	Synthesis of the composites using constituents resulted from spent cathodes						
Project partner number	1	2	3	4	5	6	Total
Project partner short name	NIMP	CITST	INFN	UM6P	UBMA	UCA	
Person-months per applicant:	3	3	6	2	2	5	21
Objectives: <i>The main objective is the preparation of: a) the composites based on a_1) (CP)_d/CN, (CP)_d and PVDF; a_2) (CP)_d/CNO and a_3) (CP)_d /Biochar/CN, where CP = PDPA and b) the fibers based on (CP)_d/CN, (CP)_d and PVDF.</i>							
Description of work (possibly broken down into tasks) and role of applicants <i>This work package will be coordinated by CITST.</i> Task 2.1 <i>Synthesis of composites: PDPA/CN/PVDF and PDPA/OCN; in this task, composites will be prepared using the following three types of procedures: a) the mixture of PVDF, (CP)_u and the (CP)_s/CN composite will be interacted with H₂SO₄, in order to obtain a mixture of PVDF, (CP)_d and the (CP)_d/CN composite; b) the in-situ oxidative polymerization of DPA in the presence of CNs in order to obtain the (CP)_d/CN composite; c) the in-situ oxidative polymerization of DPA in the presence of CNO in order to obtain the (CP)_d/CNO composite; Results will be reported in D2.1a. NIMP will lead this task with participation of INFN.</i> Task 2.2 <i>The preparation of fibers using PDPA/CN/PVDF; in this task, using the electrospinning process of the mixture of PVDF, (CP)_d and the (CP)_d/CN composite, where CP is PDPA, fibers of the type PDPA/CN/PVDF will be obtained; Results will be reported in D2.1b. CITST will lead this task with participation of NIMP.</i> Task 2.3 <i>Synthesis of PDPA/Biochar/CN composite; in this task, development of different strategies to improve their physico-chemical properties in nanocomposite/nanohybrid conception using chemical and hydrothermal doping process, chemical polymerization of monomer in the presence of different kind of CN, Biochar (in-situ polymerization and growth) will be carried out in order to obtain improved nanocomposites and/or nanohybrids containing CPs, CNs & Biochar. Results will be reported in D2.1c. UBMA will lead this task with participation of NIMP.</i> Task 2.4 <i>Characterization of composites; in this task, the physical and electrochemical characterization of the composites obtained in Tasks 2.1, 2.2 & 2.3 will be carried out. Various techniques such as: Raman scattering, FTIR spectroscopy, SEM, TEM, contact angles, thermogravimetry, XPS spectroscopy, CV, etc. will be used. Results will be reported in D2.2. INFN will lead this task with participation of all partners.</i>							
Deliverables (brief description and month of delivery) D2.1a <i>Report on the synthesis of composites: PDPA/CN/PVDF and PDPA/OCN will be focalized on the conditions of synthesis methods and the yield of polymerization reactions will be highlighted.</i> D2.1b <i>Report on the preparation of the PDPA/CN/PVDF fibers, in which experimental condition to obtain the best fibers will be detailed.</i> D2.1c <i>Report on the synthesis of composites PDPA/Biochar/CN will be focalized on the conditions of synthesis methods.</i> D2.2 <i>Report on the physico-chemical properties of composites will be focalized on the physical and electrochemical characterization of the composites by Raman scattering, FTIR spectroscopy, SEM, TEM, contact angles, thermogravimetry, XPS spectroscopy, CV, etc. All these information will be summarized into the product sheets of each composite. This is a milestone of the project labeled as M2.1.</i>							



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Work package number	3	Start date or starting event: 9				End data: 19	
Work package title	Demonstration of the performance of the composites as electrodes in RLIBs and SCs						
Project partner number	1	2	3	4	5	6	Total
Project partner short name	NIMP	CITST	INFN	UM6P	UBMA	UCA	
Person-months per applicant:	5	3	3	6	2	4	23
Objectives : <i>The main objective is the demonstration of the performance of composites prepared in the WP2 as active material for the electrodes of RLIBs and SCs.</i>							
Description of work (possibly broken down into tasks) and role of applicants <i>This work package is coordinated by UM6P.</i> Task 3.1 <i>Testing electrodes having the PDPA/CN/PVDF, PDPA/CN, PDPA/OCN & PDPA/Biochar/CN composites as active materials in SCs; in this task, the activity will be focused on testing of the electrodes in the symmetrical SC cells geometry (e.g. (-)PDPA/CN // PDPA/CN (+), etc.) and the assessing of the capacitance as function on the weight of the active compound in the electrode mass, the weight of each electrode, potential range, electrolyte type (e.g. H₂SO₄, etc.) and current densities applied during the charge-discharge tests as well as hours number in which takes place the discharge of SCs cell. The specific capacitance, energy density and power density will be evaluated taking into account the galvanostatic charge/discharge tests. The testing of the wetting properties of composites in various electrolytes will be also analyzed. A characterization of symmetrical SC cells will be also carried out by CV. Results will be reported in D3.1a. NIMP will lead this task with participation of UBMA, CITST, IFIN.</i> Task 3.2 <i>Testing electrodes containing the PDPA/CN/PVDF, PDPA/CN and PDPA/OCN composites in RLIBs; in this task, the activity will be focused on testing of the electrodes as cathodes in RLIB cells (e.g. -) Li // the PDPA/CN composite (+), etc.) and the assessing of the specific capacity as function on the weight of the active compound in the electrode mass, the weight of each electrode, potential range, electrolyte type (e.g. LiPF₆, etc.) and current densities applied during the charge-discharge tests as well as hours number in which takes place the discharge of RLIBs. The specific capacity, energy density and power density will be evaluated taking into account the galvanostatic charge/discharge tests. Results will be reported in D3.1b. UM6P will lead this task with participation of NIMP, CITST, UBMA and UCA.</i>							
Deliverables (brief description and month of delivery) D3.1a <i>Report on the electrode performance in SCs &RLIBs will be focalized on the studies on galvanostatic charge-discharge measurements, CV and/or electrochemical impedance spectroscopy of the cells using electrodes having the PDPA/CN/PVDF, PDPA/CN, PDPA/OCN & PDPA/Biochar/CN composites as active materials. This is a milestone of the project labeled as M3.1.</i> D3.1b <i>Report on the electrode performance in SCs &RLIBs will be focalized on the studies on galvanostatic charge-discharge measurements, CV and/or electrochemical impedance spectroscopy of the cells using electrodes having the PDPA/CN/PVDF, PDPA/CN, PDPA/OCN & PDPA/Biochar/CN composites as active materials. This is a milestone of the project labeled as M3.1.</i>							



Work package number	4	Start date or starting event:18				End date: 24	
Work package title	Optimization of SCs & RLIBs						
Project partner number	1	2	3	4	5	6	Total
Project partner short name	NIMP	CITST	INFN	UM6P	UBMA	UCA	
Person-months per applicant:	5	2	3	2	2	2	16
Objectives: The main objective consists in the optimization of SCs & RLIBs.							
Description of work (possibly broken down into tasks) and role of applicants							
This work package is coordinated by UBMA.							
Task 4.1 Optimization of electrodes in SC cells; in this task, the activity will involve optimization of electrodes containing composites prepared in this project in order to obtain the values of the specific capacitance of SCs equal or superior to those reported in the case composites of the type PDPA/SWNTs for the same number of galvanostatic charge-discharge cycles; the discussion on the Ragone plot will be also reported. The testing of the asymmetrical configuration of SCs (e.g. (-) PDPA // PDPA/CN (+), etc.) and the assessing of the specific capacitance will be discussed in comparison with the symmetrical configuration of SCs (e.g. (-) PDPA/CN // PDPA/CN (+), etc.). The testing of the wetting properties of composites in various electrolytes will be also analyzed. The variation of the weight of active material in the electrodes mass is envisaged, too. An increase of the doping degree of PDPA in the composites prepared in this project is also envisaged to be carried out by additional protonic acid reactions in order to test these composites in SCs cells. Results will be reported in D4.1a. UBMA will lead this task with participation of NIMP, CITST, INFN.							
Task 4.2 Optimization of electrodes in RLIBs; in this task, the activity will involve optimization of electrodes containing composites prepared in this project in order to obtain the values of the specific capacity of RLIBs equal or superior to those reported in the case composites of the type PDPA/SWNTs for the same number of galvanostatic charge-discharge cycles and current density; in this order, the variation of the weight of active material in the electrodes mass is envisaged, too; the elaboration of the reaction mechanism that takes place during charge/discharge processes in the case of the composites prepared in this project is envisaged to be reported, too. Results will be reported in D4.1b. NIMP will lead this task with participation of UM6P, CITST, UCA.							
Deliverables (brief description and month of delivery)							
D4.1a Report on optimized SCs will show the main characteristics of each SCs cells, which use the composites prepared in this project as electrode active material, i.e. specific discharge capacitance, specific energy and specific power at a fixed current density. This is a milestone of the project labeled as M4.1.							
D4.1b Report on optimized RLIBs will show specific capacity in the case of each RLIB which has composites prepared in this project in the composition of cathode. This is a milestone of the project labeled as M4.1							



Work package number	5	Start date or starting event: 1				End date: 24	
Work package title	Dissemination of results						
Project partner number	1	2	3	4	5	6	Total
Project partner short name	NIMP	CITST	INFN	UM6P	UBMA	UCA	
Person-months per applicant:	2	2	14	1	1	4	24
Objectives: <i>The objective of this WP is a) the dissemination of results by their publication as articles in ISI journals, b) communication of results at national/international conference as well as during the workshop organized by this consortium and in the close meeting, and c) the reporting of the results on the project website.</i>							
Description of work (possibly broken down into tasks) and role of applicants <i>This work package is coordinated by INFN.</i> Task 5.1 <i>Publication of min. 5 ISI articles & 6 contributions at the national/ international conferences; training for young researchers concerning the composites synthesis & their characterization. In the frame of this task, the results obtained in this project will be published min. 5 articles in ISI journals (e.g. Electrochimica Acta, Electrochemical Communications, Journal of Power Sources, Scientific Reports, Applied Surface Science, etc.) and 6 contributions at the national/international conferences (e.g. 2024 Springer EMRS, 2024 EMRS Fall Meetings, International Conference on Diamond and Carbon Materials 2024, Annual Meeting of the International Society of Electrochemistry in 2024&2025, 247th ECS Meeting) will be reported. Self-archiving in repositories and/or Open Access publishing will be considered. Trainings for young researchers from Africa concerning the composites synthesis & their characterization, the electrospinning process, the separation methods of constituents from spent cathodes of RLIBs as well as the GCD measuremntns in SCs and RLIBs will be also organized. This task will be reported in D5.1a and D5.1b. INFN will lead this task with participation of all partners.</i> Task 5.2 <i>Organization of: a) a kick-off meeting and b) a close meeting. In the frame of this task, the kick-off meeting will be organized in M1-M2, while the close meeting will be organized in the M24, both being organized by NIMP. During kick-off meeting, the collaboration agreement will be signed of all partners. During close meeting, after the presentation of the final report of this project, the perspectives concerning the consolidation of the collaboration between the partner teams in other Horizon 2022 projects will be discussed. This task will be reported in D5.1a and D5.1b. NIMP will lead this task with participation of all partners.</i> Task 5.3 <i>Workshop in Romania concerning the results obtained. This workshop will be focused onto the results obtained by six teams of this project. In the frame of workshop, will be discussed the knowledge transfer between the partners from EU, i.e. Romania and Italy, and AU, i.e. Morocco and Algeria. CITST will lead this task with participation of all partners.</i> Task 5.4 <i>A project website. This task involves the generation of a project website, which will show the information about teams involved, budget, WPs, the events (e.g., kick-off meeting, workshop, etc.). This web page will be continuously updated with the results obtained in this project. Results will be reported in D5.1a and D5.1b. NIMP will lead this task with participation of all partners.</i>							
Deliverables (brief description and month of delivery) D5.1a <i>Report on the result dissemination will show min. 2 ISI articles and min. 2 communications at the national/international conferences and the update of the web page of the project with the results obtained during the implementation of project. This is a milestone of the project labeled as M5.1a.</i> D5.1b <i>Report on the result dissemination will show min. 3 ISI articles and min. 4 communications at the national/international conferences and the update of the web page of the project with the results obtained during the implementation of project. This is a milestone of the project labeled as M5.1b.</i>							



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Work package number	6	Start date or starting event: 18				End date:24	
Work package title	Technological transfer						
Project partner number	1	2	3	4	5	6	Total
Project partner short name	NIMP	CITST	INFN	UM6P	UBMA	UCA	
Person-months per applicant:	1	2	6	1	1	1	12
Objectives: <i>The main objective is the preparation of a business plan for future exploitation and written of a patent application considering the results obtained in the frame of this project.</i>							
Description of work (possibly broken down into tasks) and role of applicants <i>Task 6.1 Business plan for future exploitation; in this task will be highlighted the opportunities to continue the cooperation between partners of this project and the identification of the business perspectives in the field of the management of the wasters coming from the energy storage field as well as the development of new energy storage devices such as SCs and RLIBs both in Europe and Africa. In this task, a strong interaction of the consortium with the two SMEs, i.e. Qi technologies and RAIT88 SRL, that are currently collaborating with Prof. S. Bellucci and who have sent letters of interest for this project will take place. Results will be reported in D6.1. This task will be led by CITST with the participation of INFN and UCA.</i> <i>Task 6.2 One patent application. In the frame of this task will be taken the decision the type of the patent application, i.e. EPO or requested in other EU countries. An example for the topic of this patent application is the performance of supercapacitors using recyclable materials from spent RLIBs. Results will be reported in D6.2. This task will be led by NIMP with the participation of all partners.</i>							
Deliverables (brief description and month of delivery) <i>D6.1 Business plan for future exploitation will show the business perspective in the field of the management of the wasters coming from the energy storage field and development of new energy storage devices and new materials for the electrodes of SCs and RLIBs. This is a milestone of the project labeled as M6.1.</i> <i>D6.2 A patent application. This is a milestone of the project labeled as M6.2.</i>							



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Work package number	7	Start date or starting event: 1				End date:24	
Work package title	Management						
Project partner number	1	2	3	4	5	6	Total
Project partner short name	NIMP	CITST	INFN	UM6P	UBMA	UCA	
Person-months per applicant:	1	1	4	2	1	2	11
Objectives: <i>To ensure that the activities are carried out and objectives are achieved within the projected time schedule and budget, and the tasks of the project are performed in high quality through an efficient partnership.</i>							
Description of work (possibly broken down into tasks) and role of applicants							
Task 7.1 <i>Technical management of project. The coordinator will maintain continuous integration of the different phases of the work flow ensuring the consistency, quality and homogeneity in project development and in the achievement of the expected results, refining and updating of work plan if necessary to keep the activities of the project in line with the objectives. Report concerning the technical management of project will be shown in D7.1a and D7.1b. NIMP will lead this task with the participation of all partners.</i>							
Task 7.2 <i>“Quality management plan”, which will provide documentation & communication procedures, etc. A “Quality Management Plan” will be prepared and circulated in the 3rd month. It will include internal management rules and methodologies, document procedures, frequency and templates; communication procedures etc. The coordinator will monitor and ensure the quality of the work performed as well as the deliverables, with contribution of all participants. Report concerning the technical management of project will be shown in D7.1a and D7.1b. INFN will lead this task with the participation of all partners.</i>							
Task 7.3 <i>“Risk management plan” for to minimize the risks & to overcome the inconvenient. A “Risk and Contingency Plan” will be prepared with contribution of WP leaders in the 3rd month and it will be re-evaluated periodically at the Management Committee Meetings. It will contain procedures for checking the project activities regarding minimization of risks and overcoming barriers. Report concerning the technical management of project will be shown in D7.1a and D7.1b. UM6P will lead this task with the participation of all partners.</i>							
Task 7.4 <i>Management committee meetings. This will be organized at each 6 months in order to monitored the progress recorded in the frame of this project. Report concerning the technical management of project will be shown in D7.1a and D7.1b. NIMP will lead this task with the participation of all partners.</i>							
Deliverables (brief description and month of delivery)							
D7.1a <i>Annual report of the project will show the scientific results, project management procedures, monitoring overall project progress, assuring integration of all activities, Kick-off and periodical meetings, IPR protection and dissemination procedures and decision making. This is a milestone of the project labeled as M7.1a.</i>							
D7.1b <i>Annual report of the project will show the scientific results, monitoring overall project progress, assuring integration of all activities, periodical meetings, workshop and close meeting, and decision making. This is a milestone of the project labeled as M7.1b.</i>							



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iv) Describe any significant risks, and associated contingency plans (max. 1 page).

Risk identified	Probability	Impact	Contingency plan
<i>The weight of CNO resulted in WP1 after the separation of the constituents of spent RLIB cathodes</i>	<i>Medium</i>	<i>Low</i>	<i>Mitigate by the achievement of various oxidation process of CN resulted after the separation of the constituents of spent RLIB cathodes</i>
<i>SCs cells having electrodes containing the composites PDPA/CN/PVDF, PDPA/CN, PDPA/OCN, PDPA/Biochar/CN show a low discharge capacitance in comparison with the cell (-) the PDPA-salt/SWNT composite// the PDPA-salt/SWNT composite (+)</i>	<i>Medium</i>	<i>Medium</i>	<i>Mitigate by interaction of the composites PDPA/CN/PVDF, PDPA/CN, PDPA/OCN with various acids (e.g. H₂SO₄, H₃PMo₁₂O₄₀, etc.). This interaction will lead to an increase of the dopage degree of PDPA and its conductivity. The variation of the ratio between the constituents of the composite materials is also envisaged.</i>
<i>RLIBs having electrodes containing the composites PDPA/CN/PVDF, PDPA/CN, PDPA/OCN, PDPA/Biochar/CN show a low capacity in comparison with the cell (-) Li// PDPA-salt/SWNT composite (+)</i>	<i>Medium</i>	<i>Medium</i>	<i>Mitigate by the interaction of the composites PDPA/CN/PVDF, PDPA/CN, PDPA/OCN with various acids (e.g. H₂SO₄, H₃PMo₁₂O₄₀, etc.). This interaction will lead to an increase of the dopage degree of PDPA and its conductivity. The variation of the ratio between the constituents of the composite materials is also envisaged.</i>
<i>Optimization of SCs cell in order to obtain high discharge capacitance</i>	<i>Medium</i>	<i>Medium</i>	<i>Mitigate by testing of electrodes prepared in this project both in symmetrical and asymmetrical SCs cells as well as the variation of the active material weight in the mass of the two electrodes will be carried out.</i>
<i>Optimization of RLIBc in order to obtain high capacity</i>	<i>Medium</i>	<i>Medium</i>	<i>The variation of the active material weight in the mass of the cathodes will be carried out</i>
<i>Dissemination of results by the physical participation at the national/international conference, kick-off meeting, closing meeting, trainings for young researchers and workshop, as a consequence of the geopolitical situation and/or the pandemic situation</i>	<i>Medium</i>	<i>Medium</i>	<i>Mitigate by the choice of the national/international conference which are organized in hybrid regim (including on-line regim). In the case of the kick-off meeting, closing meeting, trainings for young researchers and the steering committees and workshop, NIMP will considered the organization of events in on-line regime (video conferences)</i>
<i>Delay in purchasing the chemicals</i>	<i>Low</i>	<i>Low</i>	<i>All institutions have a department that are in charge with these activities and taking advantage of their experience will avoid this kind of risks.</i>



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b) Impact (max. 4 pages)

Expected outcomes and expected impacts – contribution to the expected impacts at Africa and/or international level of the relevant topic and sub-topic

Currently, worsening fossil energy depletion and global warming has made it imperative to develop not only more advanced energy storage devices but also more clean ones to control the rapid environment deterioration/pollution and to help explore more ways for clean and renewable energy [33, 34]. Among all energy storage devices, supercapacitors (SCs) with remarkable performance as high-power density, ultrafast charge/discharge rate, safety, long cycle life and eco-friendly nature, in comparison to batteries, can contribute significantly in controlling global warming offering a promising approach to meet the increasing power demands of energy storage systems in current society [35, 36]. Many African nations count on coal, oil and natural gas to supply most of their energy needs, especially in peri-urban and rural areas, but reliance on fossil fuels presents a big problem. RCLIB project aligns with the R&I partnership priorities mentioned in the AU-EU roadmap for jointly funded projects related to Climate change and sustainable energy (CCSE). The project will have an impact on improving the quality of life of rural communities in remote and isolated regions of Africa.

Overall project impact - The current proposal is fostering close collaboration between researchers and innovators by gathering a multidisciplinary consortium of industrial (CITST) and academic (NIMP, INFN, UM6P, UBMA, UCA) partners from EU (3 partners) and Africa (3 partners). The project builds on the partners' expertise and previous research in innovation projects as outlined in section c and attached key persons CVs. The proposed methods which recycle the constituents of spent RLIB cathodes into composite materials for new electrodes to be used in SCs and RLIBs will contribute to the development of novel technologies for environmentally sound recycling capacities of lithium-ion batteries in both Africa and the EU. In view of developing an exploitation strategy of project results, CITST will also explore how to promote income generation and creation of new jobs after the completion of the project (see WP6, Task 6.1). The RCLIB project takes into account the net impact on the need for jobs in energy storage and nanomaterials as well as on the economies of EU and AU countries where metals and minerals used in the manufacture of Li^+ ion batteries are mined.

Multi Annual Roadmaps - The huge progress of Lithium-ion batteries production and global sales, needs an uninterrupted supply chain of critical raw materials. At the same time, the EoL battery components induce a new environmental challenge, requiring their collection, separation of constituents and recycling/reuse to avoid further environmental pollution. The development of recycling solutions of spent Lithium-ion batteries, that contain valuable chemical elements but also harmful ones, will address both (i) the **negative impact of e-waste on the environment**; and (ii) the **shortage of the strategic metals** needed for the future development of these two technologies. In this context, the RLIB project aligns with the R&I partnership priorities outlined in the AU-EU roadmap for jointly funded projects related to Climate change and sustainable energy (CCSE). RLIB is also in line with the objectives of the LEAP RE program (MAR) and the criteria highlighted in CCSE roadmap as it focuses mainly on Capacity for joint research as the involved European and African partners bring complementary expertise to conduct high quality joint research. The overall project impact is expected to span short to long time frames over multiple levels ranging from individuals (e.g., PhDs, researchers, employees) to the African and European societies (e.g., long term economic and environmental benefits, policy incentives).



<i>Time frame</i>	<i>Impact level and size</i>	<i>MAR2 related impact</i>
<i>Short term</i>	<i>Scientific impact</i>	- New methods for cathodes recycling and reuse of compounds other than the traditional metallurgic processing for Li recycling; - Production of novel composite materials; - Preparation of electrodes for batteries and supercapacitors via conventional methods and electrospinning.
<i>Short term</i>	<i>Capacity building</i>	- Closer collaboration between researchers, innovators and funders; - Improved professional competences and expertise (including R&D, managerial, business, communication) of the team member; - Closer institutional cooperation between Africa and Europe.
<i>Medium term</i>	<i>Societal and policy impact</i>	- Creation of jobs through use and reuse of EOL components; - Creation of policy incentives to encourage collection and reuse of rechargeable Li batteries for materials other than the traditional Li recovery; - Increased awareness for the economic and environmental effects involved with collection and reuse of Li batteries.
<i>Medium term</i>	<i>Technological impact</i>	- Recycling technology will become more efficient through uptake of the project's results; - Manufacturing of new cathodes for batteries and SC.
<i>Medium term</i>	<i>Business</i>	- The SME involved in the project will exploit the project results and the identified markets both locally and through the collaboration of the industrial partners who offered support letters (e.g., specialized electrospun fibers, IPR)
<i>Long term</i>	<i>Economic</i>	- Reduced production costs and increased recovery rates; - Transition from a linear to a circular economy.
<i>Long term</i>	<i>Environmental impact</i>	- Reduced materials used for new products; - Incentive for increased collection of Li batteries; - Decrease in heavy metals and hazardous chemicals waste.
<i>Long term</i>	<i>Business</i>	- Increased economic gain through the uptake by main consumers of Li batteries and SC: automotive, marine, industrial, and power.

Stakeholder impact: At individual level, the close collaboration between researchers, engineers and business-oriented stakeholders (CITST) will have a long-term impact on their expertise while opening new avenues for future collaborations. At institutional level, initial steps towards a future commercial exploitation in WP6 will have an economic impact on both the involved SME and on potential commercialization through spin-offs. All partners will strengthen its national and international collaboration and mutual understanding towards new avenues for common research initiatives and high impact publications. In Morocco and Algeria, the acquired expertise will aid UM6P, UCA and UBMA partners to become involved in the avenues opened by the countries' natural resources which kept growing in recent years. Special attention will be given to a high exchange of scientific knowledge between the parties involved by achievement of trainings focused on the nanomaterials used as electrode active materials in SCs and RLIBs, separation methods of constituents from spent cathodes of RLIB, electrospinning techniques for African specialists.



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Gender balance One of the sectors where women participation needs a boost is the energy sector, as stated by Evrydiki Fekka, Head of Lithium Batteries Product Management at Sunlight Group Energy Storage Systems [37]. Consequently, the project is favoring women participation as leaders of the consortium partners (3 out of 6, including the coordinator). At team level, the percentage of women is also prevailing which helps in promoting women in science, engineering, technology and business.

Dissemination of the project results

The whole consortium will be involved in a sustained dissemination effort to ensure visibility and awareness through various dissemination channels as outlined in the table below. In this way, the results of the project can be broadly anchored, new cooperation partners and opportunities can be activated, and innovative research projects can be initiated on a national and international level with particular focus on collaboration with African countries.

	1	2	3	4	5	6
	General public	Industry	Policy makers	Public authorities	Academia	Media
Project Website and website of the partner organizations (updated regularly)	X	X	X	X	X	X
Brochure, flyer to be distributed at fairs and conferences (e.g., Re-Battery, TechWeek, E-Waste World)		X	X	X	X	
Press releases at least yearly. Regular press work is carried out via ISO's own press distribution list and via the Science Information Service (idw)	X	X	X	X	X	X
Social media (LinkedIn, Twitter, etc.) updated regularly (at least 1 post per month)	X	X	X	X	X	X
Newsletter by each partner at least yearly		X	X	X	X	
Scientific publications in journals: min. 5 articles in ISI journals (e.g. Electrochimica Acta, Electrochemical Communications, Journal of Power Sources, Scientific Reports, Applied Surface Science, Journal de la SOACHIM West African Association of Chemistry Journal, etc.);		X			X	
Participation in forums and exhibitions, e.g. Re-Battery, TechWeek, Battery Recycling in Frankfurt Messe, etc.		X	X	X	X	X
Presentations at conferences: minimum 6 participations at conference e.g., 2024 Springer EMRS, 2024 EMRS Fall Meetings, International Conference on Diamond and Carbon Materials 2024, Annual Meeting of the International Society of Electrochemistry in 2024&2025, 247th ECS Meeting, Annual conference of West African Association of Chemistry, etc.		X	X	X	X	
A workshop organized by the project	X	X	X	X	X	X

The dissemination plan and data management will follow the Guidelines to the Rules on Open Access to Scientific Publications and Open Access to Research Data in Horizon 2020¹³ and the HORIZON 2020 Online Manual.¹⁴ The aim is to implement transparency and reproducibility of the research performed within the project.

Intellectual property rights (IPR) management

All partners have agreed to follow the general IPR rules of the H2020 program¹⁵ as will be defined in the Consortium Agreement (CA). All partners will continuously contribute to the identification of project

¹³ https://ec.europa.eu/research/participants/data/ref/h2020/grants_manual/hi/oa_pilot/h2020-hi-oa-pilot-guide_en.pdf

¹⁴ https://ec.europa.eu/research/participants/docs/h2020-funding-guide/cross-cutting-issues/open-access-data-management/open-access_en.htm

¹⁵ Your Guide to IP in Horizon 2020 ([Link](#))



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results that may qualify for IPR protection. The partners will designate one amongst them, or a third party, as Project IPR manager to: raise IPR awareness, highlighting patent opportunities, etc.

Ownership of background is unaffected by the project and will be identified in the CA. It will be granted non-exclusively, free-of-charge for the duration of the project to the partners. A non-disclosure agreement will be concluded if valuable background information needs to be made available.

*Regarding **foreground** (including information, materials and knowledge whether or not these can be protected), each party owns the resulting IPR generated by it within the project. Patentable results shall be identified and filed by the party that developed them at its own cost or transferred to another partner under fair and reasonable conditions. For joint results and inventions, the partners will gain joint ownership, in which case they agree who shall be responsible for protecting and maintaining rights.*

*For the **commercial exploitation**, IPR management is important to ensure clear ownership rights in order to be able to sell the product after project completion. In this respect, we will discuss already in the beginning of the project different IPR strategies in order to decide on the applicable rules that prevent misunderstandings at later stage. Researchers will have the right to publish research results, subject to the terms of consortium agreement, with a reasonable period of notification of publication.*



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c) Quality and efficiency of the implementation (max. 2 page)

The differences and complementarity to max. 5 other projects carried out by the consortium partners in the past three years can be highlighted as follows: **i) NIMP** has implemented the project “Development of doped vanadium oxide/graphene composites for ultra-performance batteries and supercapacitors by physical vapor deposition for sustainable and eco-friendly energy storage application (Project ID: PN-III-P2-2.1-PED-2019-4519), implementation period: 23.10.2020-22.10.2022; **ii) CITST** was involved in the implementation of the project “Technological innovation for nano-sized and uniform structures through enhanced control of the electrospinning process” (Project IT: PN-III-P2-2.1-PED-2019-4021), implementation period 11.11.2020-11.11.2022; **iii) Erasmus+ KA107- 2019-2021 -UBMA-UNAMUR,2014-2016- South Africa – Algeria: Scientific Research & Technological Development Cooperation (Synthesis Of Metal-Oxide/Graphene foam (GF)-MO/CNTs Hybrid Nanostructured Materials & Their Applications in Energy Related Areas); iv) UCA & UM6P** were involved in SESA Project (H2020-LC-GD-2-3-2020) entitled Smart Energy Solution for Africa (2022-2024). SESA’s objective is to mitigate climate change & avoid lock-in situations while improving access to sustainable energy under affordable and reliable conditions. UCA & UM6P are concerned with three living Labs (rural & urban & Battery Recycling). The aim of Battery recycling is to implement activities & technologies that will minimize resource use & waste production and reduce emissions by slowing down, reducing and closing energy and material cycles. This will be achieved by durable construction, maintenance, repair, re-use and recycling of individual components, particularly focusing on batteries. These activities are in good relation with the WP1 & WP3 of RCLIB Proposal, **v) Prof I. SAADOUNE (UM6P)** is leading the LEAP-RE RESTART project (Horizon 2020, Grant Agreement 963530, 2022-2024). The objective of this project is to implement a full value chain for recycling EoL LiBs & PV, shifting from linear economy to circular economy, thus reducing waste disposal as well as minimizing dependence on important primary materials. The scheduled activities in RESTART project will be useful for implementing the WPs of RCLIB. The expertise of partners and their complementarity in this consortium can be quantified as follows: **i) NIMP’s team (Romania)** has expertise in: a) synthesis and physico-chemical properties of a_1) CPs like as polyaniline, PDPA, poly(pyrrole), poly(3, 4-ethylenedioxythiophene), poly(N-vinyl carbazole) and son on; a_2) the composites based on CPs and single-walled carbon nanotubes, a_3) the ternary composites of the type poly(o-phenylenediamine), poly(vinylene fluoride) and double-wall carbon nanotubes; and a_4) purification/separation of SWNTs; b) advanced characterization techniques such as Raman scattering-SERS, IR spectroscopy, UV-VIS-NIR spectroscopy, photoluminescence, XRD, SEM, HRTEM, TGA, XPS, CV, electrochemical impedance spectroscopy (EIS), dielectric spectroscopy, contact angles and c) charge-discharge galvanostatic tests for RLIB and SC. The leader of NIMP’s team is Dr. M. Baibarac (**F**). Relevant ISI articles for the project topic: [1,29,38-43] (**topic: SC cells**),[2](**topic: RLIB**). A profile address: ORCID:0000-0003-1107-514X, Researcher ID:B-5731-2011. **The assigned tasks** are:1.1,1.2,1.3,1.4,2.1,2.2,2.3,2.4,3.1,3.2,4.1,4.2,5.1,5.2, 5.3,5.4,6.2,7.1,7.2,7.3,7.4. **Efforts:** a) personnel consist in 3 senior researchers (SR), 5 researchers (R) & 7 research assistants (RA) (4 PhD students & 2 Master students); 19 person-month (PM); b) existing infrastructure: http://erris.gov.ro/CMATPHYS_ADVMAT; <http://erris.gov.ro/CEUREMAVSU>; Origaflex multichannel potentiostat/ galvanostat, OrigaLys; c) the requested budget (RB): 160 K€; **ii) CITST’s team** is a Romanian research-oriented SME, founded in 2007, focusing in the field of electrospinning on developing custom made electrospinning components (e.g., spinnerets) & materials for small scale applications. CITST’s team has expertise in electrospinning a wide range of polymers and in fine tuning the electrospinning process through additional electrodes, external electric fields and lenses in order to obtain uniform and/or directional polymer fibers. CITST has been involved in several international (2 HORIZON 2020, 12 transnational projects, 3 ERASMUS grants) and in more than 20 national grants. Relevant for this call are five grants which have specifically focused on achieving improved process control in electrospinning and on developing novel materials for sensing applications: (1) ELECTROSPIN, funded under Partnerships PNCD II-003/2007; (2) MAGNANO & TECHMAT funded under the PN-III



PED-2016 & 2019 Program; (3) two projects funded under the national program POC-A1. Dr. O. Cramariuc (F) will lead the CITST team. **The assigned tasks** are: 1.1,1.3,1.4,2.1,2.2,2.4,3.1,4.1,4.2,5.1,5.2,5.3,5.4,6.1,6.2,7.1,7.2,7.3,7.4; **Efforts**: a) the main team members are 5 SR, 3 R & 7 RA supported by the technicians of the team; 15 PM; b) existing infrastructure: <https://eiris.eu/ERIF-2000-000U-1995>; c) the RB: 90 K€; Own contribution (OC): 19 K€. iii) **UM6P's team (Morocco)** has expertise in is a leading African university in Batteries Materials development. UM6P offers numerous facilities in relation with battery assembling & testing and battery materials characterizations (XRD, SEM, BET, ICP...). The team is coordinated by Dr. I. Saadoun (M). Relevant publications: [44,45]. **The assigned tasks** are: 1.1,1.3,1.4,2.1,2.4,3.1,3.2,4.1,5.1,5.2,5.3,5.4,6.2,7.1,7.2,7.3,7.4. **Efforts**: a) the main team members are SR and young R; 16 PM; b) existing infrastructure: glove Box MBraun, 16-channels battery cyler, XRF analyzer, impedance meter, TGA, DSC; c) RB: 50 K€; OC: 56.8 K€; iv) **UBMA's team (Algeria)**-LEREC Laboratory has expertise in in : a) synthesis of a₁) different type of nanostructured carbon (MWNTs, Graphene, functionalized CNTs or Graphene, VACNTs) and amorphous carbon from biomass recycling (Biochars and their activation) and their nanocomposites /nanohybrids; a₂) Polyaniline nanofibers and their nanocomposites/nanohybrids; b) advanced characterization techniques such as Raman and FTIR spectroscopy, XPS, XRD, FESEM, HRTEM, TGA/DTA, BET; c) electrochemical measurements via CV, EIS and charge-discharge galvanostatic tests (GCD) for SC. The leader of UBMA-Algeria's team is Dr.O. Guellati (F). Relevant ISI articles for the project topic: [8] (topic: Biochar), [46-49] (topic: electroactive material for SCs), [10,48-50] (topic: doped CNs), [51,52]. A profile address: ORCID: 0000-0001-5586-7152, Researcher ID: 57189466066. **The assigned tasks** are: 1.1,1.3,1.4,2.1,2.3,2.4, 3.1,3.2,4.2,5.1,5.2,5.3, 5.4,6.2,7.1,7.2,7.3,7.4. **Efforts**: a) personnel consist in 4 SR, 7 R (5 PhD students & 2 Master students), 11 PM; b) existing infrastructure: <http://lerec.univ-annaba.dz/>; c) RB: 50K€. v) **INFN's team (Italy)** is an organization dedicated to the study of the fundamental constituents of matter. The team has an extensive expertise in nanoscience and nanotechnology, nanomaterials and nanocomposites modelling, manufacturing, characterization and nanodevices realization. The leader of INFN is Dr. Stefano Bellucci (M). He is: a) currently Professor habilitated in Theoretical Physics of Fundamental Interactions and of Condensed Matter; b) a member of the Lab. Council of INFN Laboratori Nazionali di Frascati, participating to the activities of the Council and providing scientific and managerial advice to the Laboratory Director; c) a member of the INFN-LNF Committee for organizing and managing General Seminar and Laboratory Scientific Colloquia. His research interests include theoretical physics, condensed matter, nanoscience and nanotechnology, nanocarbon based composites, biomedical applications. Role in project: INFN node will focus on 0.5 – 18 GHz measurements by Vector Network Analyzer, SEM, AFM, DC conductivity and network analyzer characterizations. Relevant publications: [26,53,54]. **The assigned tasks** are: 1.1,1.3,1.4,2.1,2.4,3.1,4.1,5.1,5.2,5.3,5.4,6.1,6.2,7.1,7.2,7.3,7.4. **Efforts**: a) personnel consist in 1 SR, 3 post-doctoral fellows and 4 graduated students; 39 PM; b) existing infrastructure: hot filament CVD equipment, thermal CVD equipment for patterned deposition of carbon nanotubes, arc discharge setup for production of high purity and low-defect CN samples, Koetterman cabinet for chemical purification and functionalization of nanocarbons, Atomic Force Microscope (AFM) (APE Research), scanning electron microscope TESCAN VegaII, endowed with Bruker microanalysis and QUANTAX 400 software of analysis, Detector STEM for the acquisition of images in brightfield and in darkfield. Agilent E5071C ENA Network Analyzer, set-up for measuring electric and magnetic permittivity in conductive samples; c) RB: 200 K€. OC: 105 K€. vi) **UCA's team (Morocco)** has expertise in: a) the electrochemical studies; and b) wasters management. Relevant articles: [55-57]. The team is coordinated by Prof. R. Hakkou (M). **The assigned tasks** are: 1.1,1.3,1.4,2.1,2.4,3.1,4.2,5.1,5.2,5.3,5.4,6.1,6.2,7.1,7.2,7.3,7.4. **Efforts**: a) the main team members: 3 SR, 1PhD student; 22PM;b) existing infrastructure: potentiostat/ galvanostat, Rigaku SmartLab X-ray diffractometer, UV-2600 Shimadzu UV-VIS spectrophotometer, etc.; c) RB: 50K€; OC: 56.8 K€.



d) References (max. 2 page)

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5. Justification of resources (max. 1 1/2 pages)

Please describe the projected costs and justify requested funds on the separate Excel sheets.

NIMP budget is equal to 160.000 €. The personnel costs are equal to 65.000€, they corresponding the salaries for 3 senior researchers, 5 researchers & 7 research assistants (4 PhD students & 2 Master students). The full salary costs comply with national regulations for each persons' qualification and take into account the foreseen involvement in terms of working time (19 PM). The R&D equipment infrastructure use costs are equal to 20.000 € and this are foreseen for devices which exceed 500 Euros (national regulations) such as the upgrading of the Origaflex potentiostat from Origalys with two channels, software licenses, etc. Costs of materials are equal to 31.667 €, these corresponding to different substances (solvent, monomer, super P carbon, etc.), liquid nitrogen, etc. Travel costs correspond to the participation to the conferences mentioned in WP5, including the registration taxes, as well as visits to partners are equal to 12.000€. Sub-contracting costs correspond to the taxes for open-access publications and these are equal to 8.000 €. Overhead is calculated as 20% of the direct costs minus equipment costs and this is equal to 23333€.

CITST budget of 109.000 € will be covered 90.000€ by the project funding and 19.000€ by the companies' own funds, according to the national funding regulations. The personnel costs of 67.000 € will cover the salaries of 5 senior researchers, 3 researchers and research assistants supported by the technicians of the team. The full salary costs comply with national regulations for each persons' qualification and take into account the foreseen involvement in terms of working time (15 PM for seniors, technicians, and young researchers). The equipment funds of 7.000 € are foreseen for devices which exceed 500 Euros (national regulations) such as laptop, additional high power voltage and liquid pump. Costs of materials are equal to 9700 €, these corresponding to the purchasing of polymers, spinnerets, solvent resistant tubes, etc.. It is important to note that the budget can be transferred between equipment and materials without any restrictions, as per national rules. Travel Budget, for the participation to the conferences mentioned in WP5 (including the registration taxes) and visits to partners, is equal to 1710 €. Sub-contracting costs correspond to the taxes for open-access publications and these are equal to 8000 €. Overhead is calculated as 20% of the direct costs minus equipment costs and this is equal to 15590€.

INFN budget of 305.000€ will be covered 200.000 € by the project funding and 56000 € by the companies' own funds, according to the national funding regulations. Personnel costs are equal to 152.000 €, they corresponding the salaries for 1 senior researcher, 3 post-doctoral fellows and 4 graduated students (27 PM). Costs of materials are equal to 77.000 €, these corresponding to the various chemical compounds, laboratory glasses, etc. Travel budget for conferences including the registration taxes is equal to 20.000 € for participation at the conferences mentioned in WP5 as well as the events organized by NIMP (e.g., kick-off meeting, close meeting, workshop, etc.). Overhead is calculated according to the national rules.

UM6P budget of 106800€ will be covered 50.000 € by the project funding and 58.000 € by the university own funds, according to the national funding regulations. Personnel costs are equal to 26000€, they corresponding the salaries for 1 full-time young researchers (10 PM) and partial salaries of permanent researchers (6PM). The R&D equipment infrastructure use costs are equal to 58.800 €, these being allocated to purchase of a battery cyler to test novel batteries built-up from recycled components. Costs of materials are equal to 20.000 €, these corresponding to the various chemical compounds, laboratory glasses, etc. Travel budget for the participation to conferences mentioned in WP5 (including the registration taxes) as well as the events organized by NIMP (e.g., kick-off meeting, close meeting, workshop, etc.) is equal to 2.000€. Overhead is 0%.

UBMA budget is equal to 50.000 €. The personnel costs are equal to 17.500 €, these covering the salaries for 4 senior researchers and 5 PhD young researchers (11 PM). Costs of materials are equal to 27.000 €, these corresponding to the various chemical compounds, laboratory glasses, Potentiostat



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accessory, etc. Travel costs are equal to 5.500€, they corresponding to the participation at conferences mentioned in WP5 as well as at the events organized by NIMP (e.g., kick-off meeting, close meeting, workshop, etc.). Overhead is 0%.

UCA budget of 106.800 € will be split with 50.000 € coming from project funding and 58.000 € coming from the university own funds, as required by national funding regulations. The budget includes personnel costs of 32.800 euros for salaries of partial salaries of permanent researchers and for young researcher (1 Ph.D. student) and 22 PM. The budget also includes 50.000 € for R&D equipment and infrastructure, such as a waste battery crusher. Materials, including various chemical compounds and laboratory glasses, will cost 19.000 €. Sub-contracting costs, or hiring outside help, will be 2.000 euros. Travel expenses for conferences and events organized by NIMP will be 3.000 euros. There will be no overhead costs.